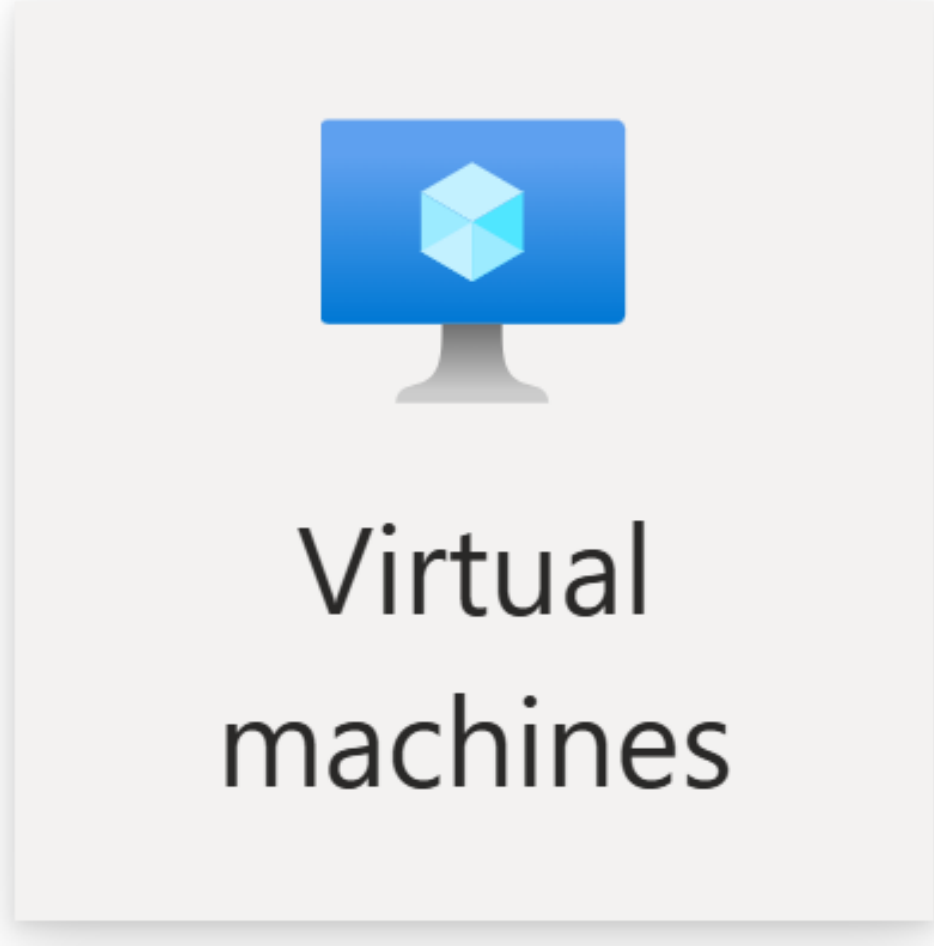
What is an Azure Virtual Machine?

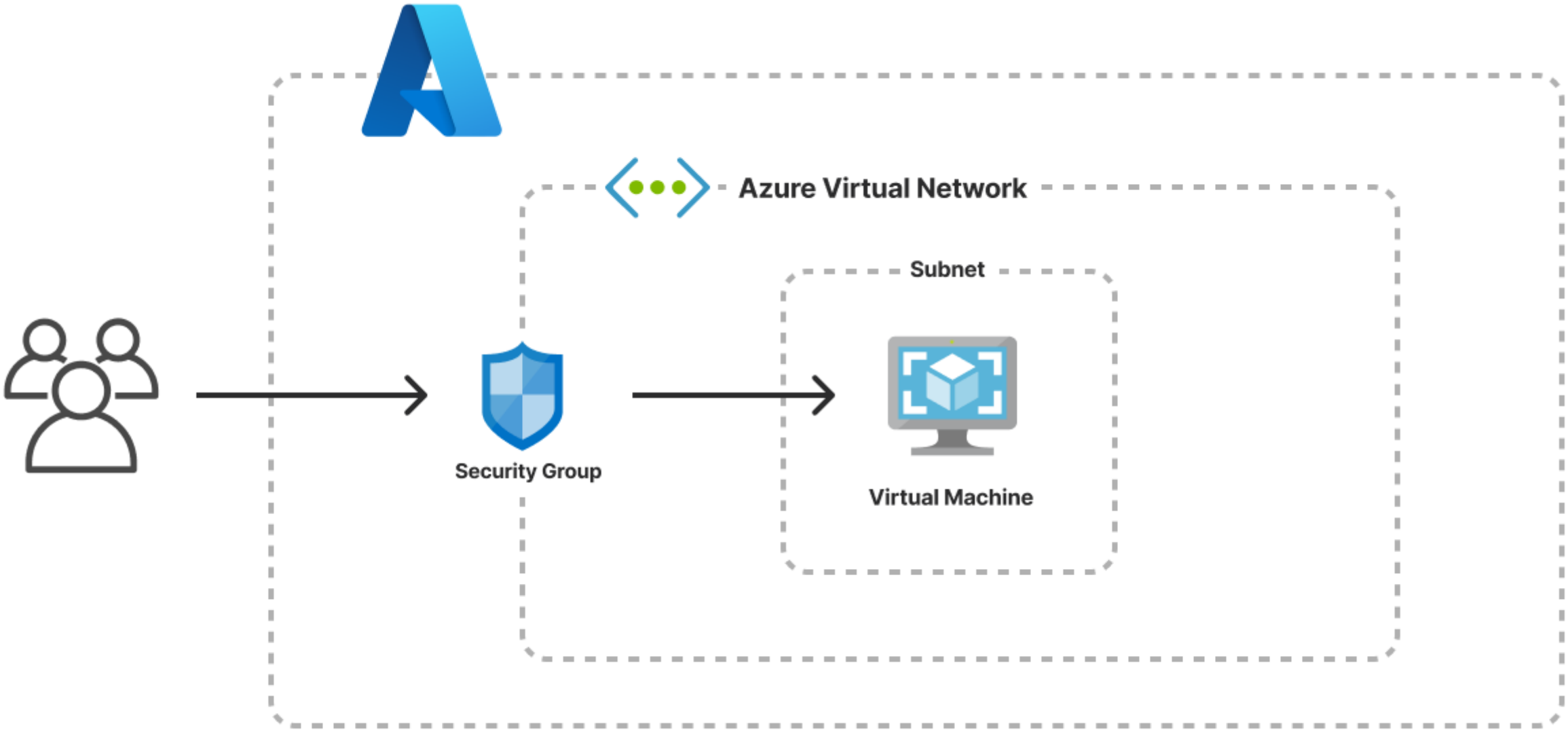
An **Azure Virtual Machine (VM)** is an **IaaS (Infrastructure as a Service)** offering where Microsoft provides:

- Physical hardware
- Networking
- Data center

And **you manage**:

- OS (Windows/Linux)
- Patches
- Applications
- Security inside VM




Azure Virtual Network
Subnet
Security Group
Virtual Machine

Azure VM Architecture

A VM is **never alone**. It always belongs to:

Component	Purpose
Resource Group	Logical container for resources
Virtual Network (VNet)	Private network
Subnet	Network segmentation
NIC	Network card for VM
OS Disk	Windows/Linux OS
Data Disk (optional)	Extra storage
Public IP (optional)	Internet access
NSG	Firewall rules

VM Types

Common VM Series:

Series	Use Case
B-Series	Low cost, burstable
D-Series	General purpose (most used)
E-Series	Memory optimized
F-Series	Compute optimized

VM Deployment Methods

You can create VM using:

1. **Azure Portal**
2. **Azure CLI**
3. **PowerShell**
4. ARM / Bicep Templates (conceptual)

📌 **AZ-104 asks:**

> Which method is best for automation?
> ✅ **ARM / CLI / PowerShell**

Hands-On: Create VM

Steps (Portal):

1. Go to **Microsoft Azure Portal**
2. Create **Resource Group**
3. Select:
   - Image (Windows Server / Ubuntu)
   - Size (B2s / D2s_v5)
4. Configure:
   - Username / Password or SSH
5. Networking:
   - VNet
   - Subnet
   - Public IP
   - NSG (allow RDP / SSH)
6. Review + Create

⏱️ VM deployment: **2–5 minutes**

6️⃣ Azure VM Login Methods

OS	Method	Port
Windows	RDP	3389
Linux	SSH	22

📌 Security Rule:

Never open **RDP/SSH to 0.0.0.0/0** in production.

VM Pricing Basics

You are charged for:

- VM **running time**
- Disk (even when VM stopped)
- Public IP
- Outbound bandwidth